

Ryan Qin

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Education

University of Waterloo

Sept 2024 - Jun 2029

Bachelor of Applied Science in **Electrical Engineering**, Honours, Co-op

Technical Skills

Languages: C, C++, C#, Assembly, Verilog, VHDL, Java, Python, HTML/CSS, JavaScript

Technologies: FPGA, STM32, DSP, FSM, Robotics, RTOS, Linux, TCP/IP, MongoDB, Node.js, React

Development Tools: Cadence, Altium, LtSpice, COMSOL, MATLAB, DSO, Fusion 360, TensorFlow, Android, AWS, Git,

Professional Experiences

HSIO Compliance Engineer | TensorFlow, OpenCL, FPGA, C++, C#, Cadence, DSO, I²C
AMD

May 2026 - Present

- Designing **FPGA-based** USB 2.0 I/O testing board, aimed to **autonomously** test Link Layer and drop/droop compliance
- Developing **STM32-based I²C** module for **automatic** HDMI **EDID** setting, helping to reduce manual rewiring work
- Optimizing electrical Compliance analysis calculations using **OpenCL**, utilizing GPU parallelism for consecutive test runs
- Implementing **USB 4.0 test-automation** software in **C#**, designed to automate instruments over **TCP/IP** protocol

Electrical Engineer | LtSpice, Altium, EPLAN, C++, TCP/IP
ATS Corporation

Sep 2025 - Dec 2025

- Managed **full-cycle** design and verification for **four parallel** electronics projects, ensuring **on-time, spec-compliant** delivery
- Developed an **inductively isolated** pulse conditioner board in **Altium**, protecting PLC against stray voltage while **reducing 80% of Sub-GHz noise** from turbocharger's hall-effect tachometer
- Developed a **TCP/IP-enabled STM32** pneumatic leak-testing control board with a **thermal flow sensor** and **customized hardware drivers**, reducing power consumption by **40%** while decreasing unit's size by **50%**
- Designed and implemented **PID control loops** and **dither circuits** for a fuel nozzle testing system's pressure regulator, enabling **25%** more accurate simulation of turbojet engine operating conditions.

Electrical Coordinator | C#, .Net, MySQL, COMSOL
WSP

Jan 2025 - May 2025

- Recognized by **WSP Canada's** Rail and Transit **VP** for OCS engineering Software's **innovation intake** potential
- Independently developed Overhead Contact System (**OCS**) engineering software in **C#**, reducing Calculation time by **40%**
- Optimized traction power substation's cooling system layout in **COMSOL FEA**, aimed to extend transformer life by **10 years**

Projects

TakeOne 🏆 | CAN-BUS, Python, C++, TCP/IP

Finalist at **Hack The North** Sep 2026

- Designed I/O architectures using isolated USB I/O and CAN Bus, ensuring reliable communication against ground lift and loop
- Architected the Robots Communication and 42V/12V Power Delivery Network, ensuring stable and reliable drivetrain system
- Implemented AI-based IK robotics arm processing
- Designed Real-Time Control loop for robotic arm

TTL/FPGA Processor 🏆 | Verilog, Vivado, RTL, FSM, FPGA, DSP, RISC-V, TTL

Dec 2025 - Jan 2026

- Implemented **pipelined vector/tensor operations** in Verilog/SystemVerilog, exploiting **FPGA DSP** slices to increase compute throughput per clock cycle
- Designed and built a **custom CPU** from **discrete TTL logic**, implementing the ALU, register files, control unit, and clocking architecture from **first principles**
- Extended the processor architecture in **Verilog** for **FPGA deployment**, adding pipelining, forwarding, memory/port mapped I/O and SPI flash

Autodump 🏆 | C++, TCP/IP, Fusion 360, Python, OpenCV

3rd place at **MakeCU** Nov 2025

- Architectures mechanical layout Using Fusion 360, allowing to rapid prototype solution without wasting materials
- Engineered an Arduino-based serial **distributed I/O** system, focused on **simplifying** software-hardware integration and **reducing PWM-signal jitter and integrity** for servo and ultrasonic sensors

- Designed **ArUco** marker navigation system in **OpenCV**, allowing the robot to **autonomously navigate** back to home position

Competition RC  | PyTorch, Fusion 360, LTSpice, AVR Assembly

Finalist at **Formula Null** May 2025

- Built an RC car with **50%** recycled material and custom 3d printed steering mechanism
- Designed a **custom** airsoft motor controller using MOSFET stages, relays, and Arduino for PWM signal conversion, achieving a **fast lap record** of **17 seconds** compared to **average 25 seconds** lap time
- Implemented a **lane-keeping algorithm**, combining **PID** and **PyTorch**-based vision to adjust steering from USB camera